

# EG1311: Design & Make

## CAD Intro

Dr. Jason S. Ku

# Why Fusion?



- Fully-featured CAD software
- Cross-Platform
- Free for education use (personal restricted)
- Close to SolidWorks in workflow

# How to get?



Fusion\_Installatio...



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## AUTODESK Fusion

Single User Install Process and  
Browser Access to Fusion



**There are two paths that students and educators can take to create an Autodesk account with educational access to Autodesk education products including Autodesk Fusion. The first path, option 1, available to students and educators.**

**Option 1**  
**Individual Access**

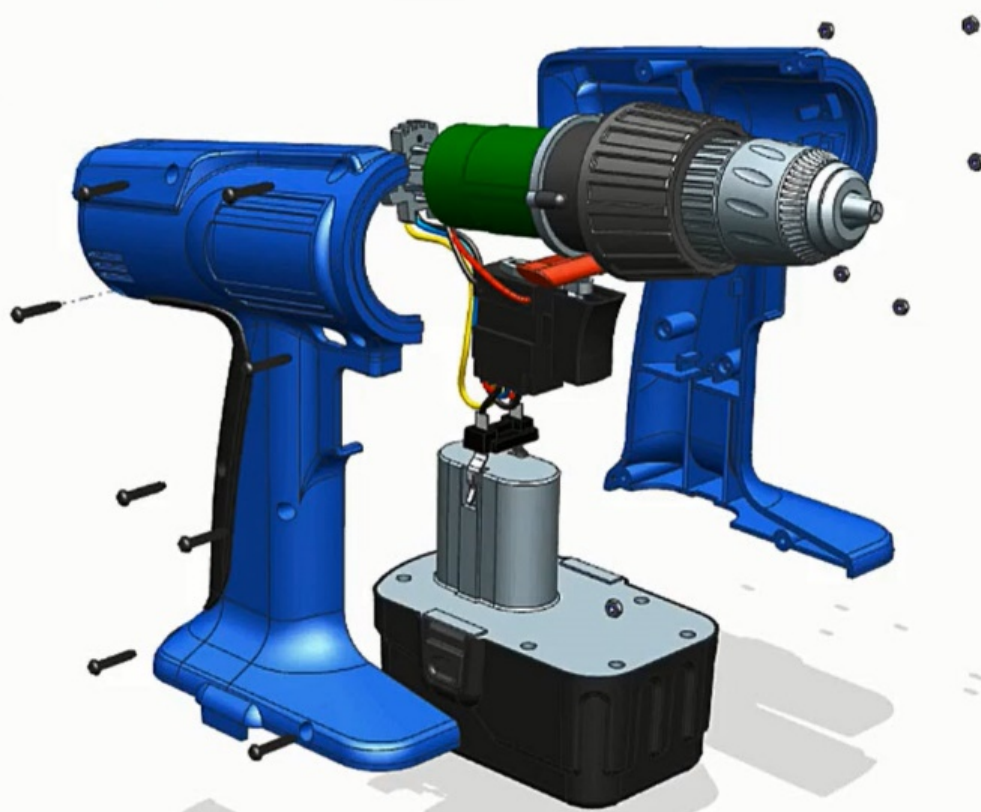
# What is Fusion?



- Software for Computer Aided Design (CAD)
- Construct 2D and 3D geometries
- Design parts and assemblies before build
- Outputs the input for tools for:
  - fabrication (3D printing, Laser cutting, etc)
  - analysis (FEM, stress, conduction, etc)
  - rendering (visualization, animation)

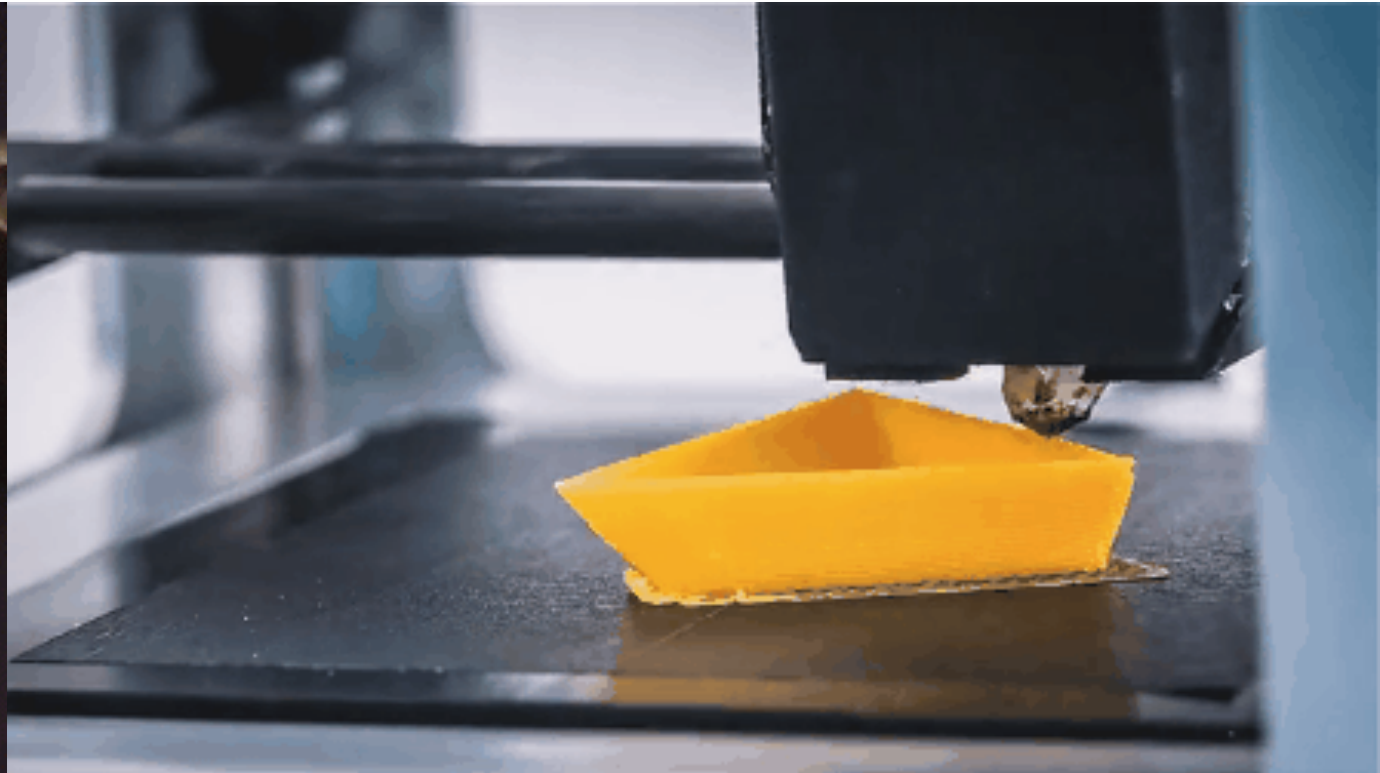
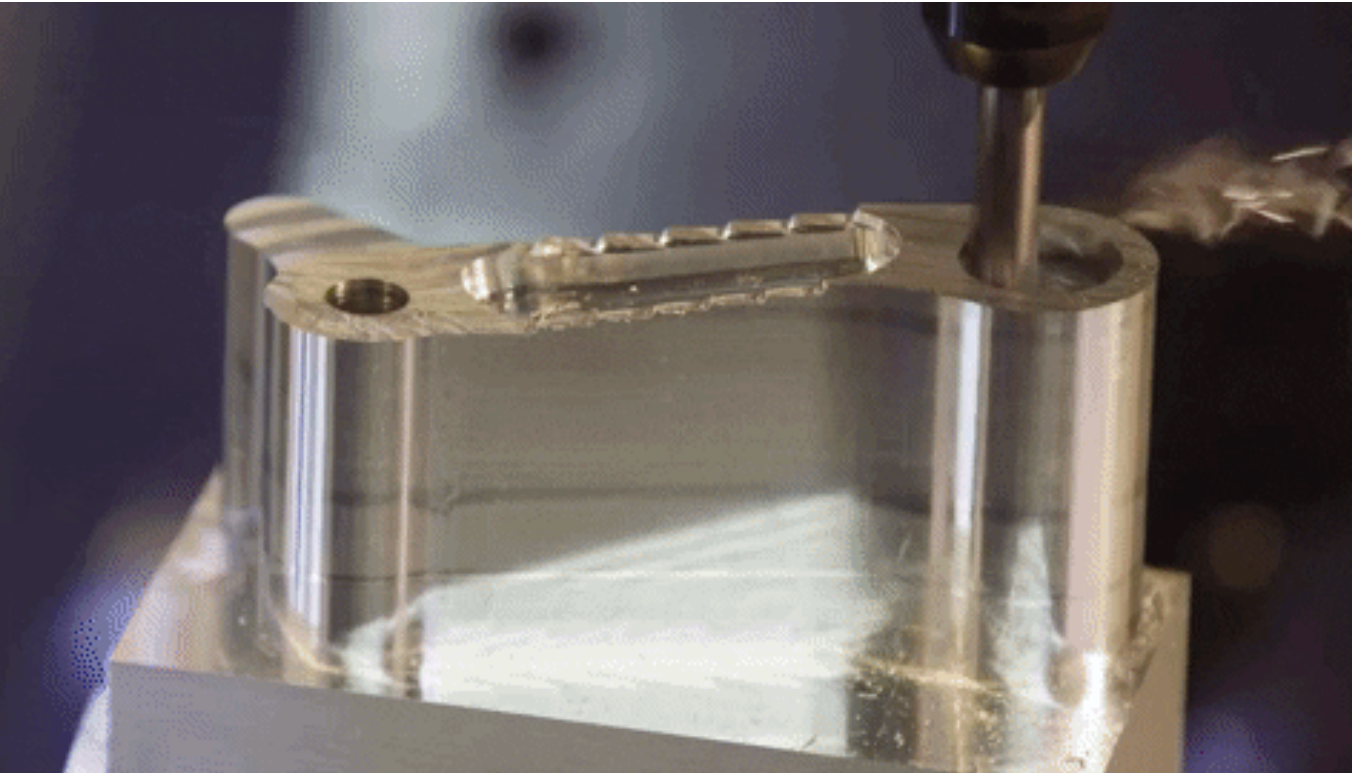


# What can it do? Design





# What can it do? Fabrication

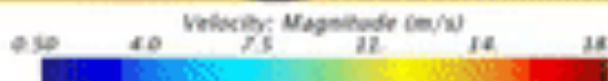
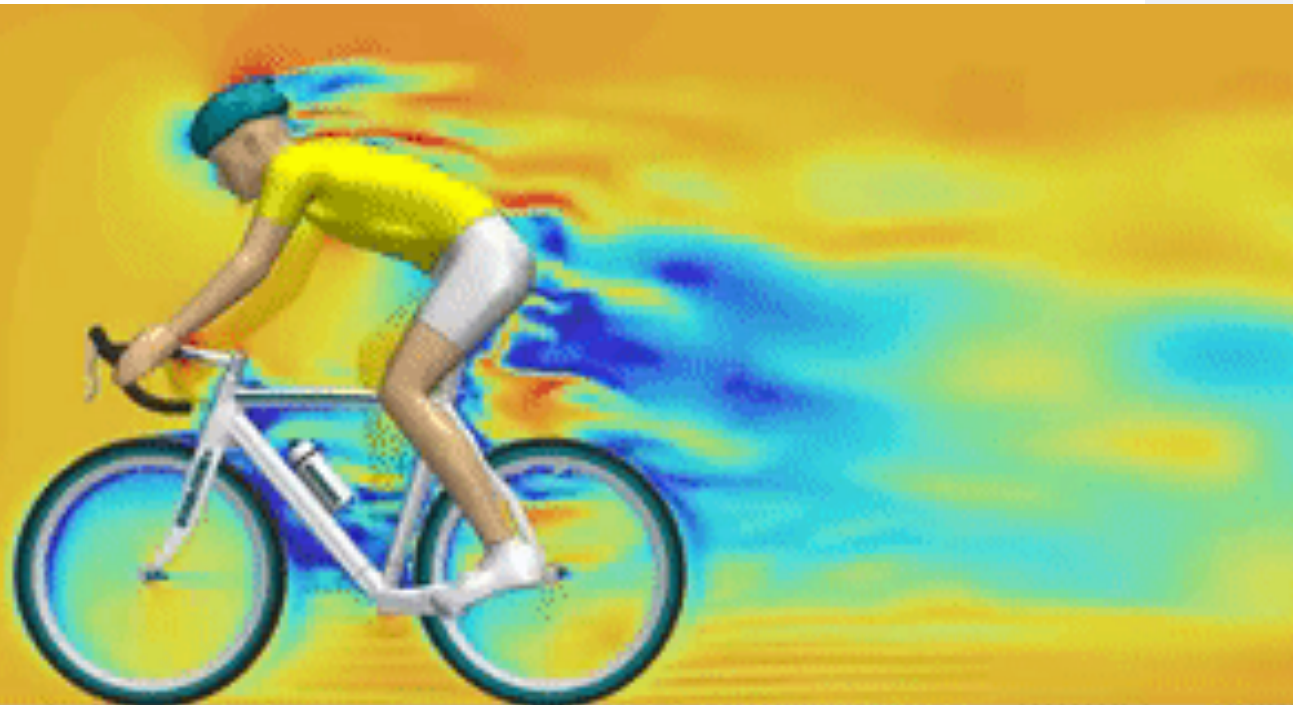
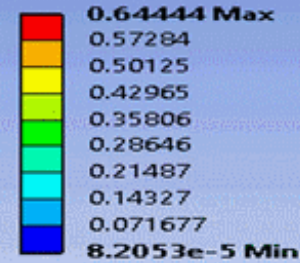




# What can it do? Analysis

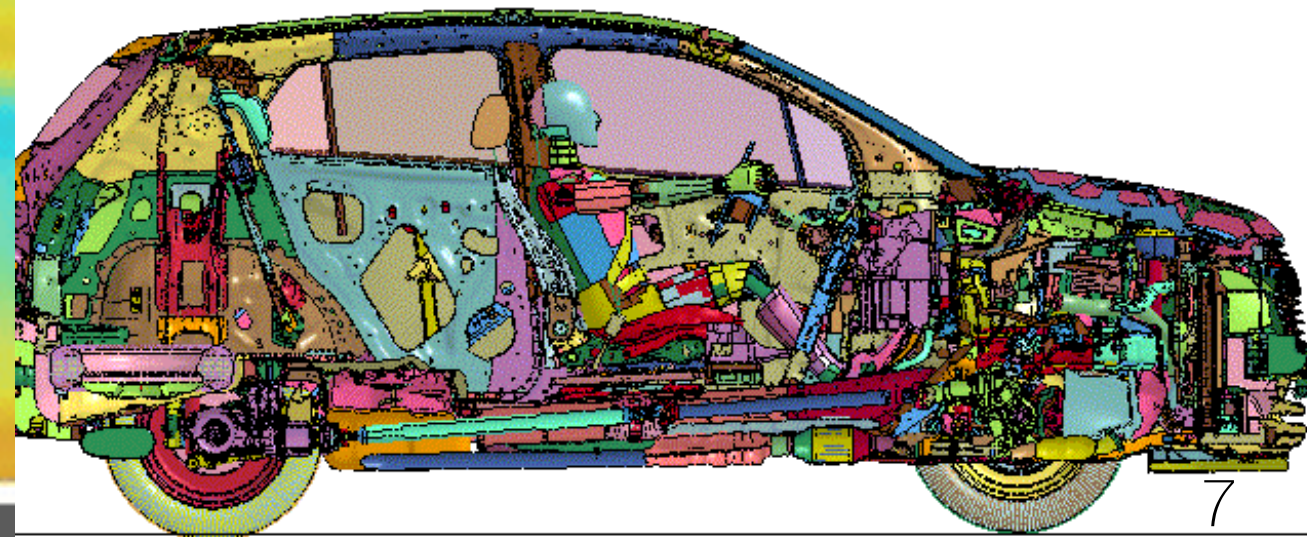


A: Static Structural  
Equivalent Stress  
Type: Equivalent (von-Mises) Stress  
Unit: MPa  
Time: 0  
1/22/2016 2:27 AM



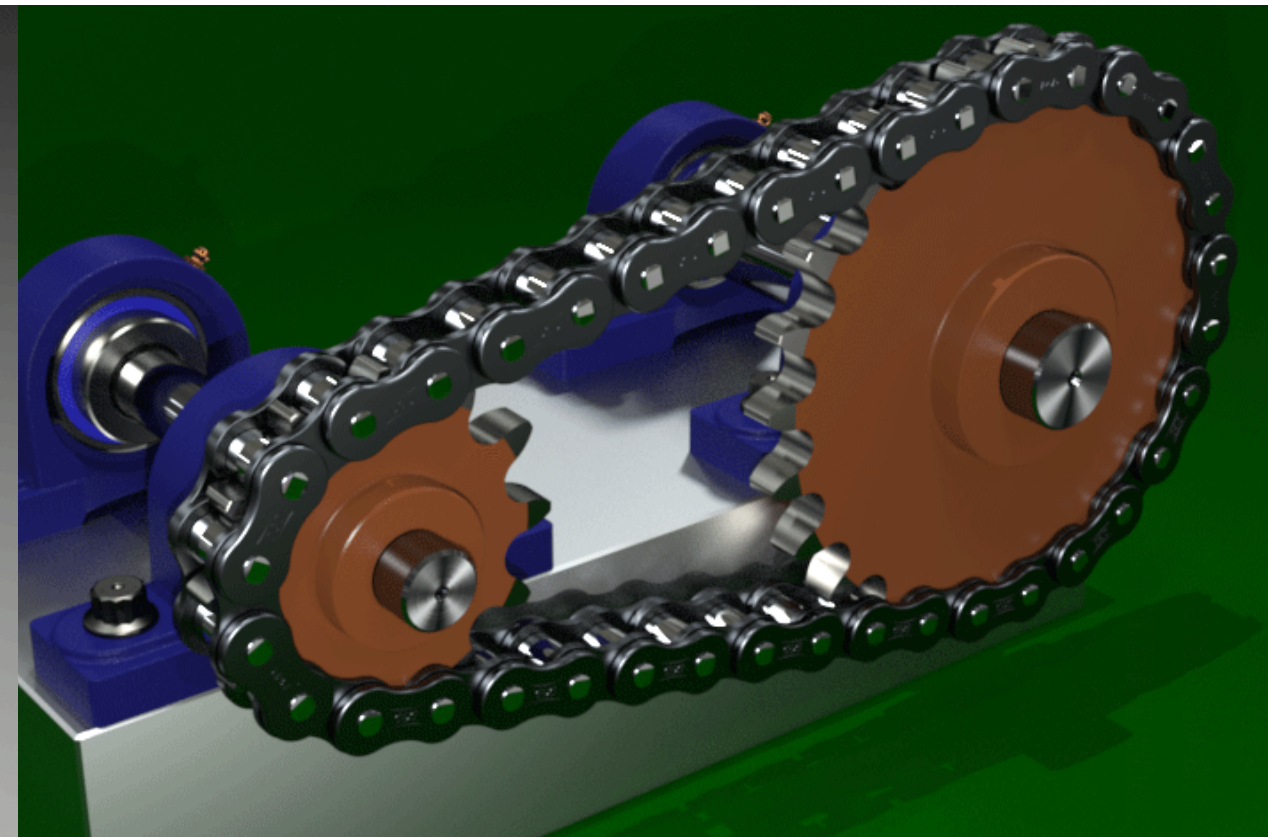
MakeAGIF.com

Time = 0  
MY15 CHEVY TRAX





# What can it do? Rendering

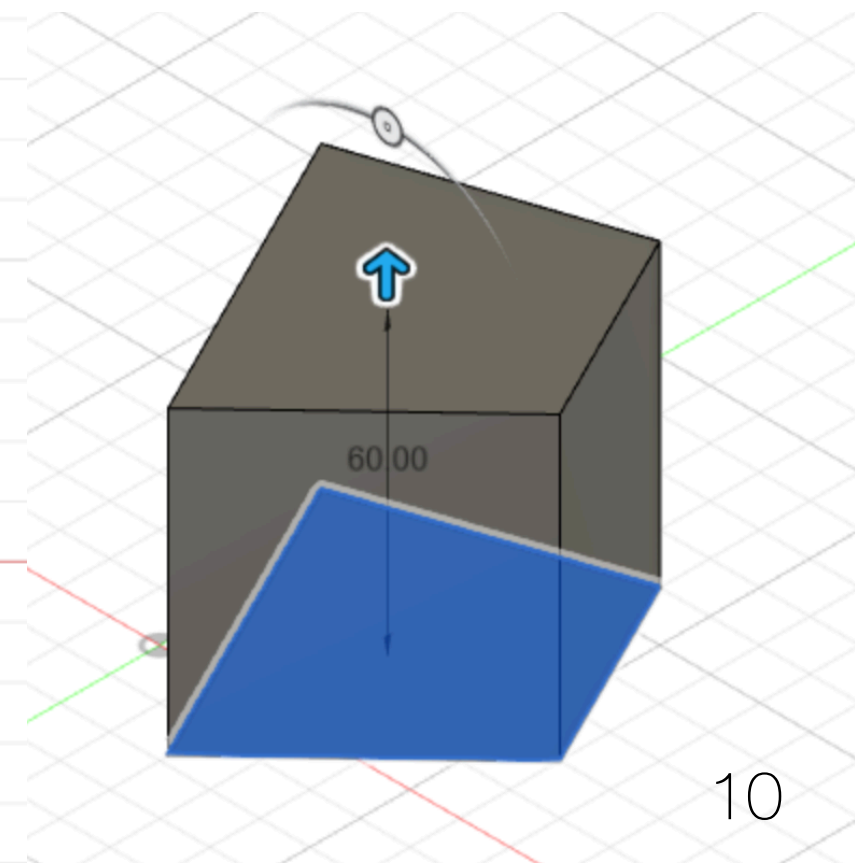
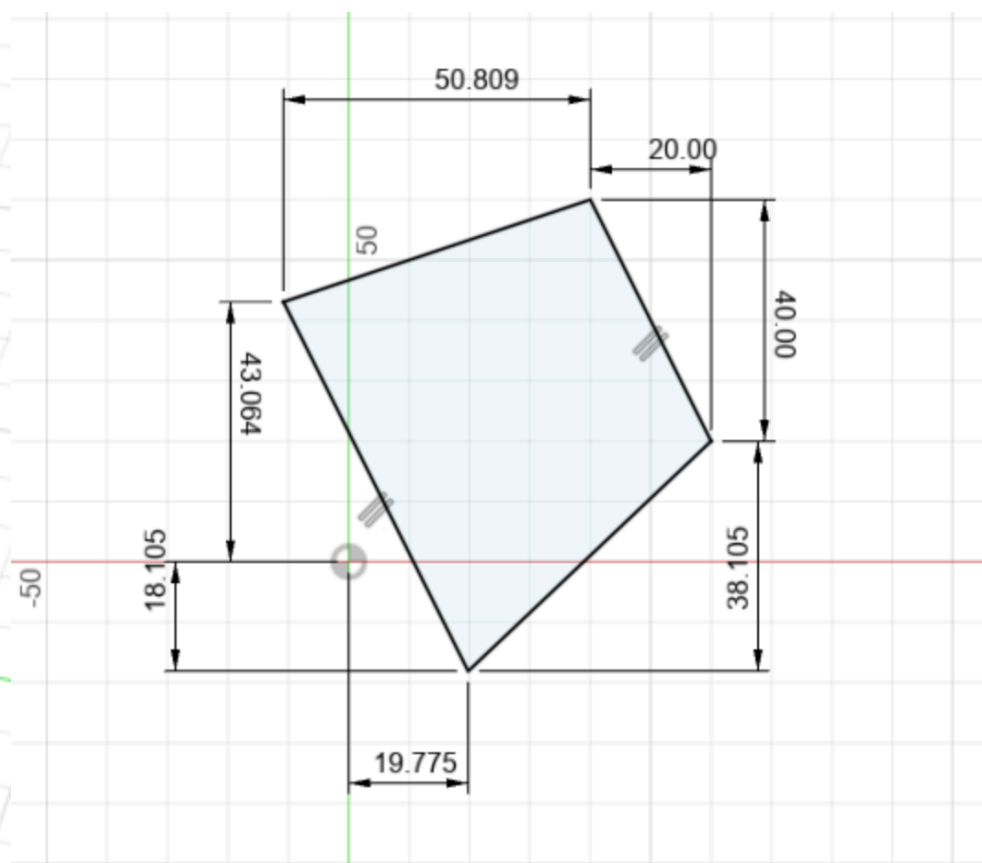
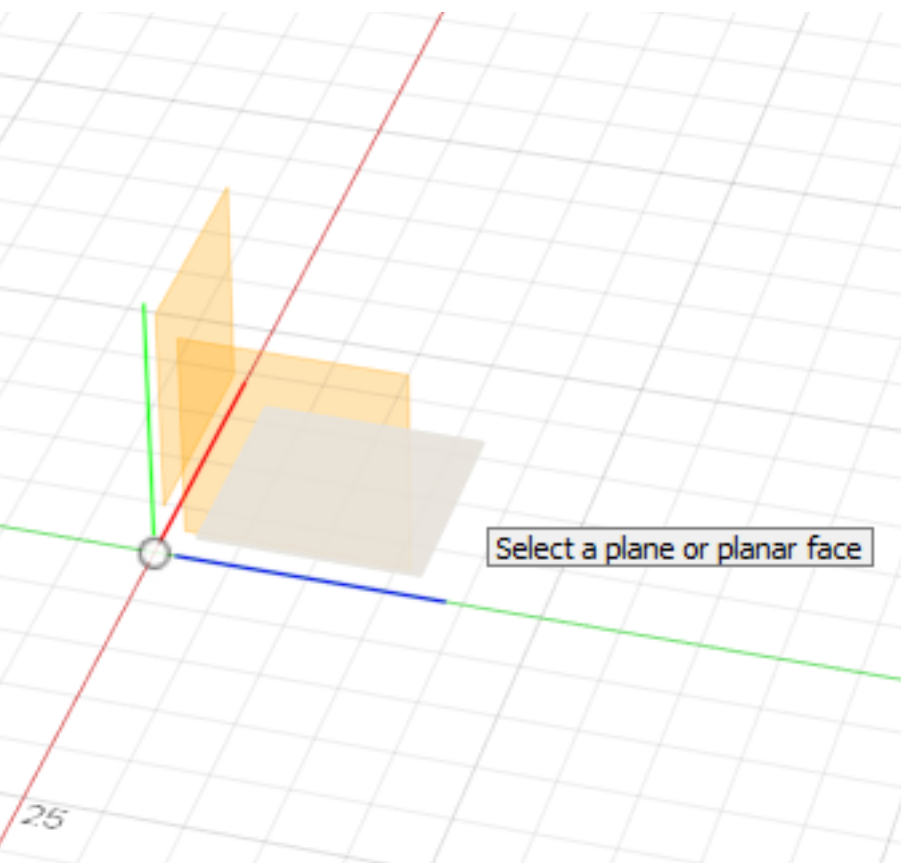


# Goals

- Basic Introduction to CAD
- Learn Fusion 360 interface
- Learn to create parametric 2D sketches
- Learn to create 3D parts
- Learn to export DXF files for laser cutting
- Learn to create multi-part assemblies
- Learn to create a dimensioned drawing
- Learn to style, animate, and measure

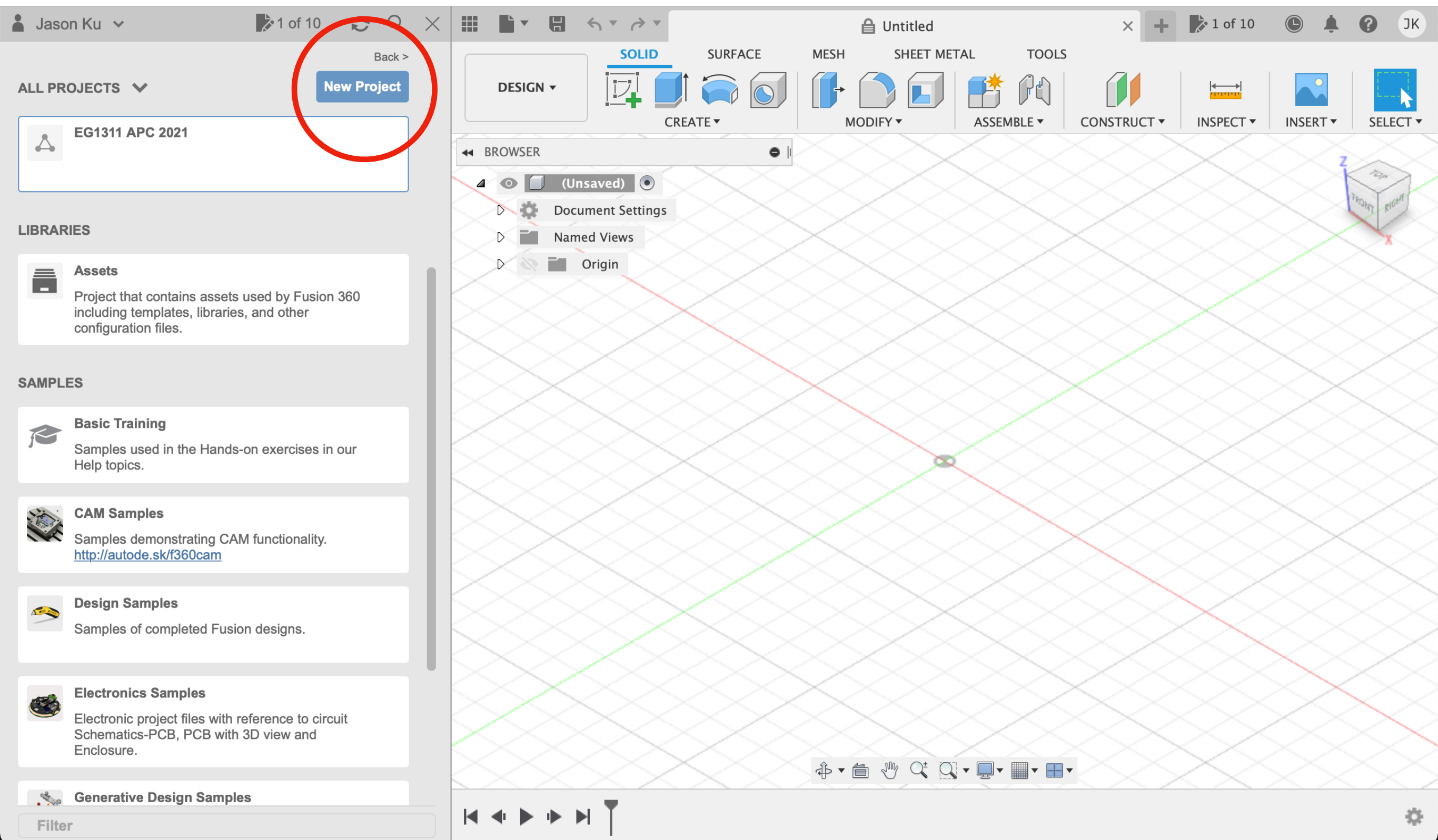
# Workflow

- 1) Select a Sketch Plane
- 2) Draw/Dimension/Constrain a 2D Sketch
- 3) Construct a 3D Solid Feature from Sketch
- 4) Repeat

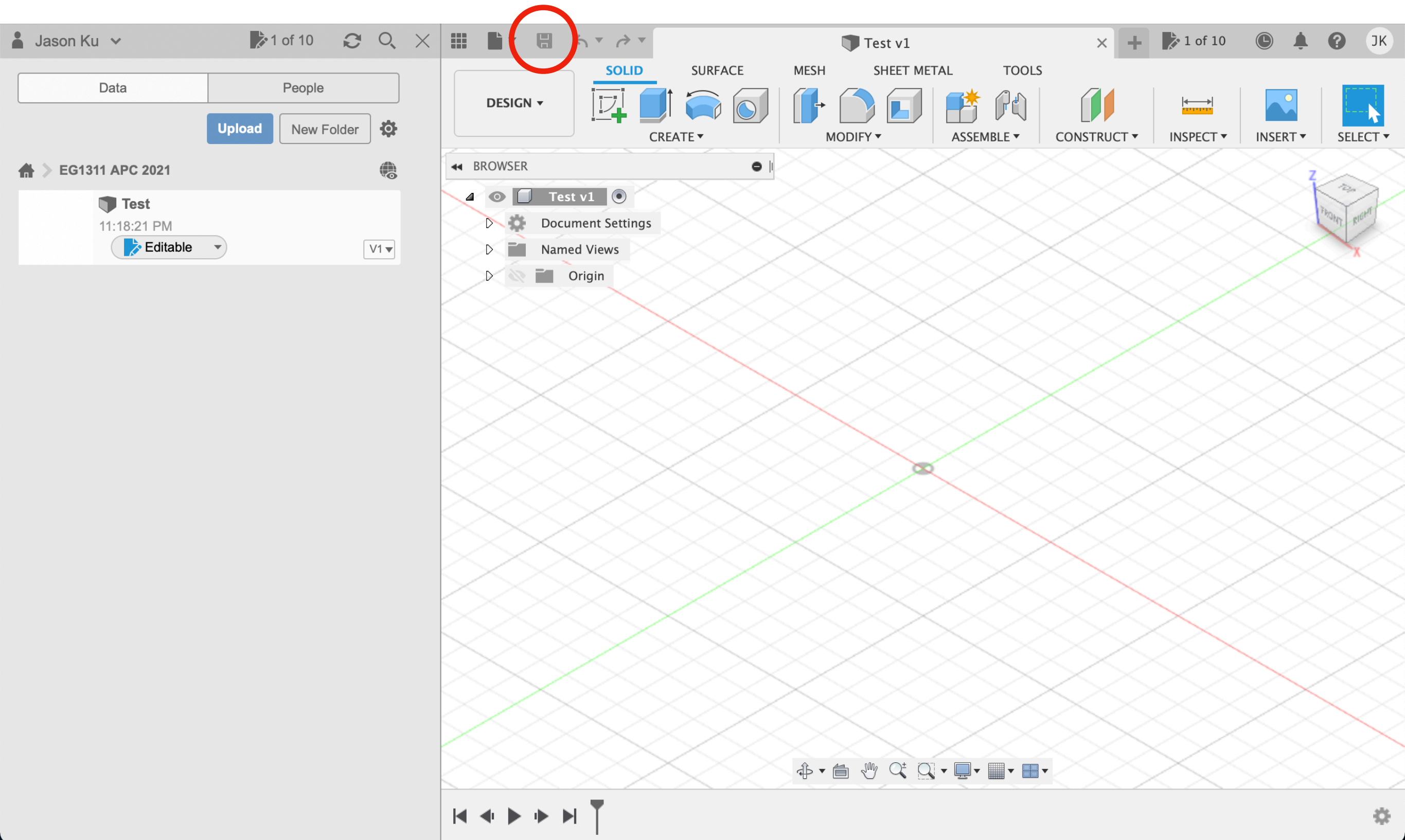




# Interface

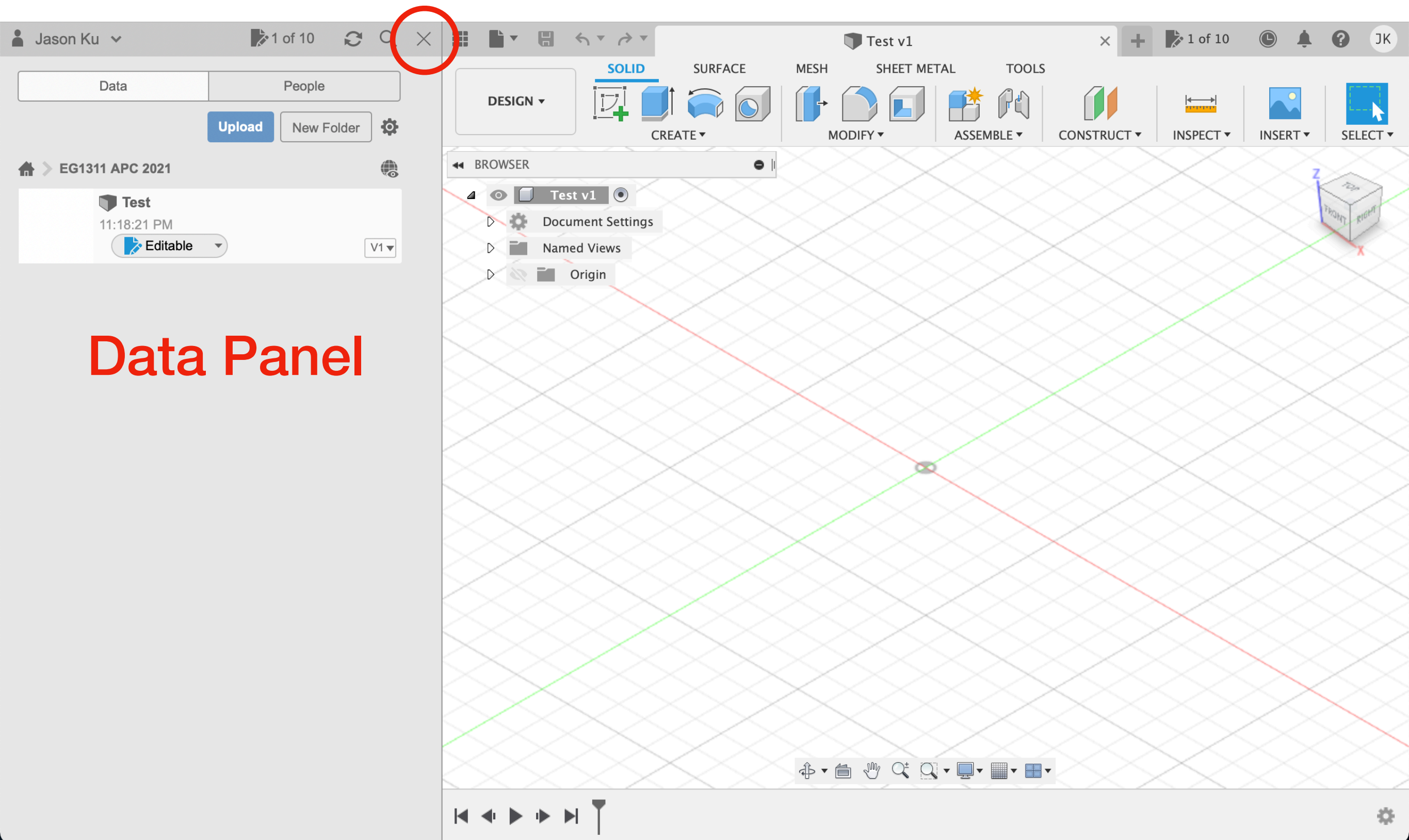


# Interface



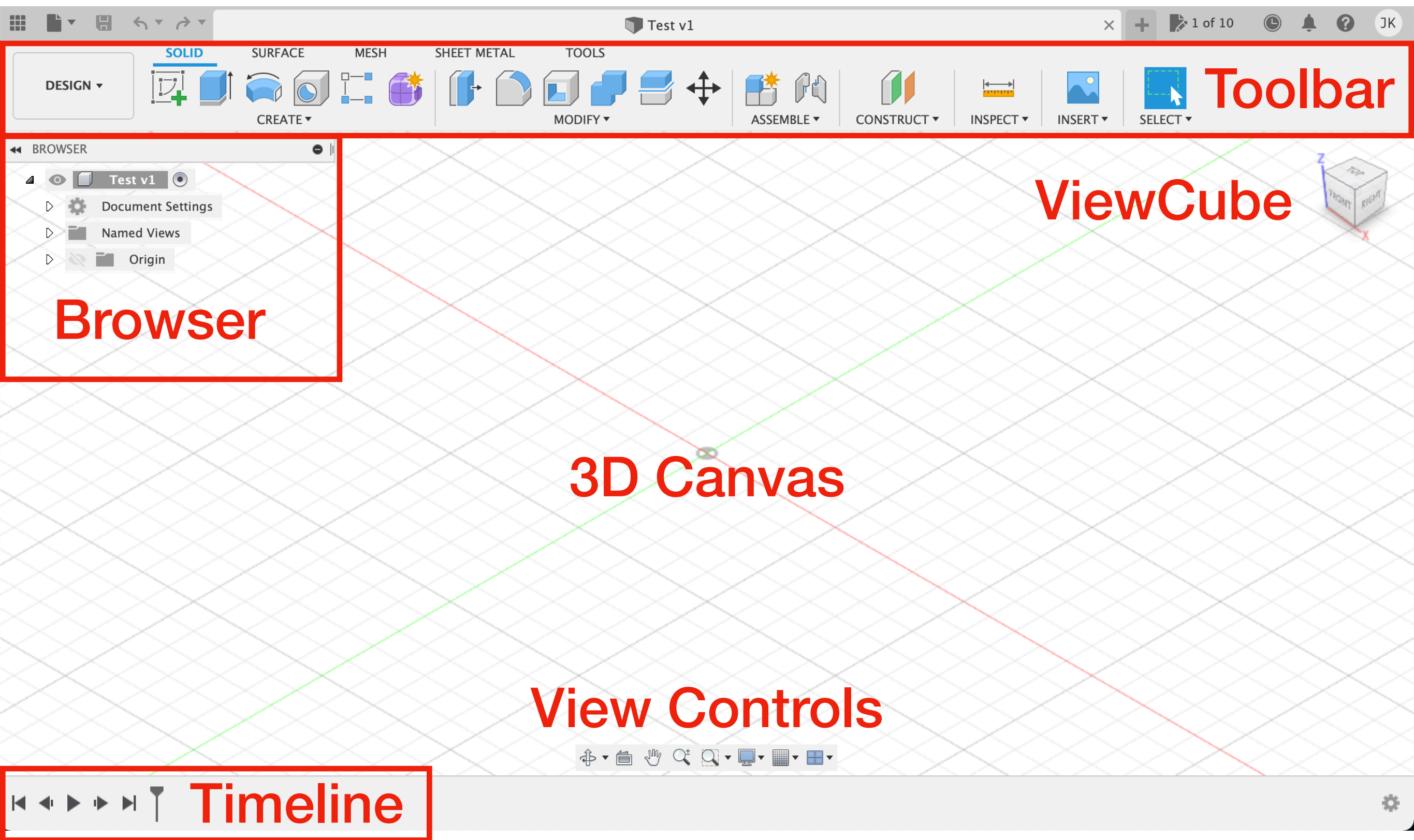


# Interface

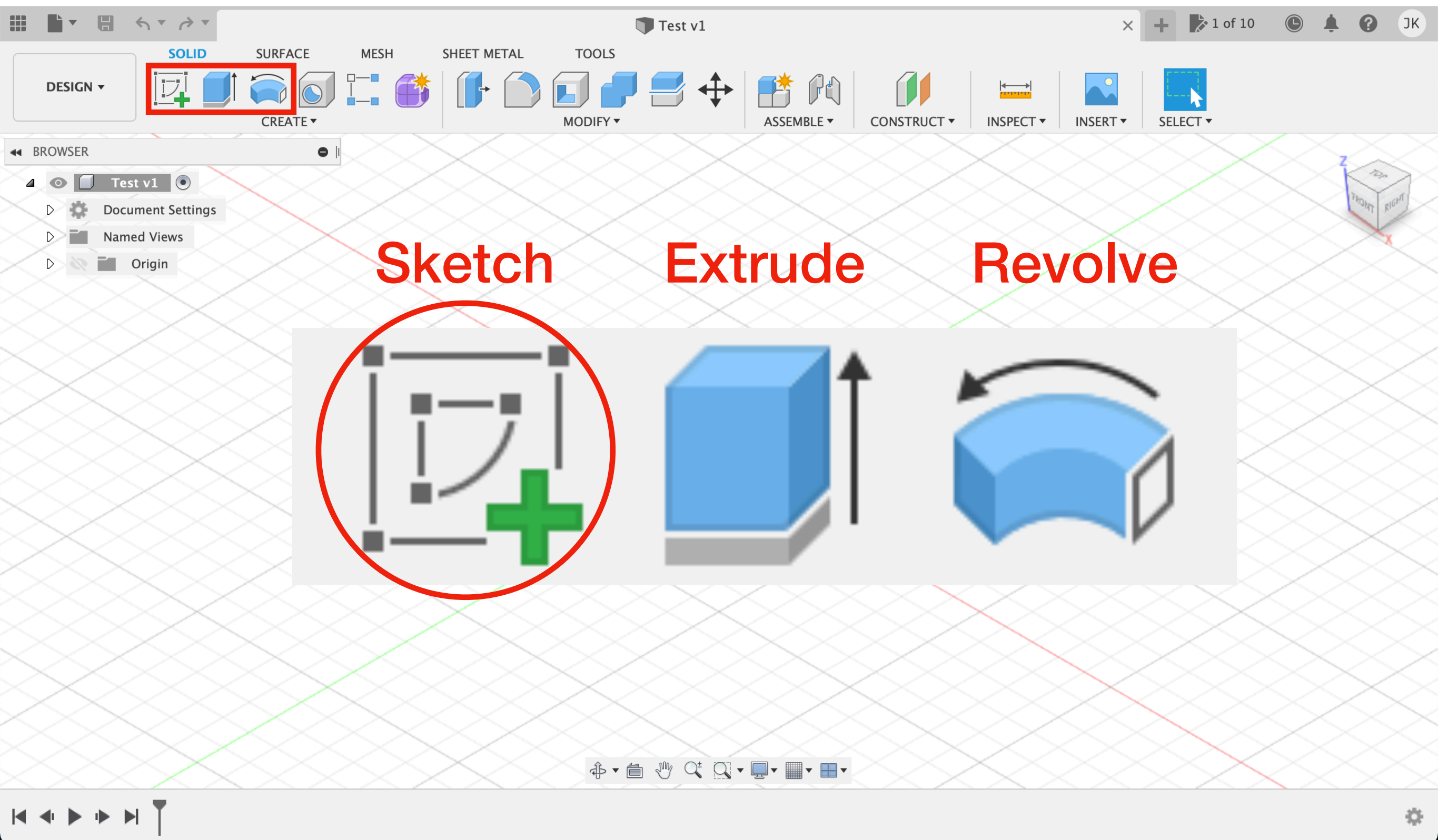


Data Panel

# Interface

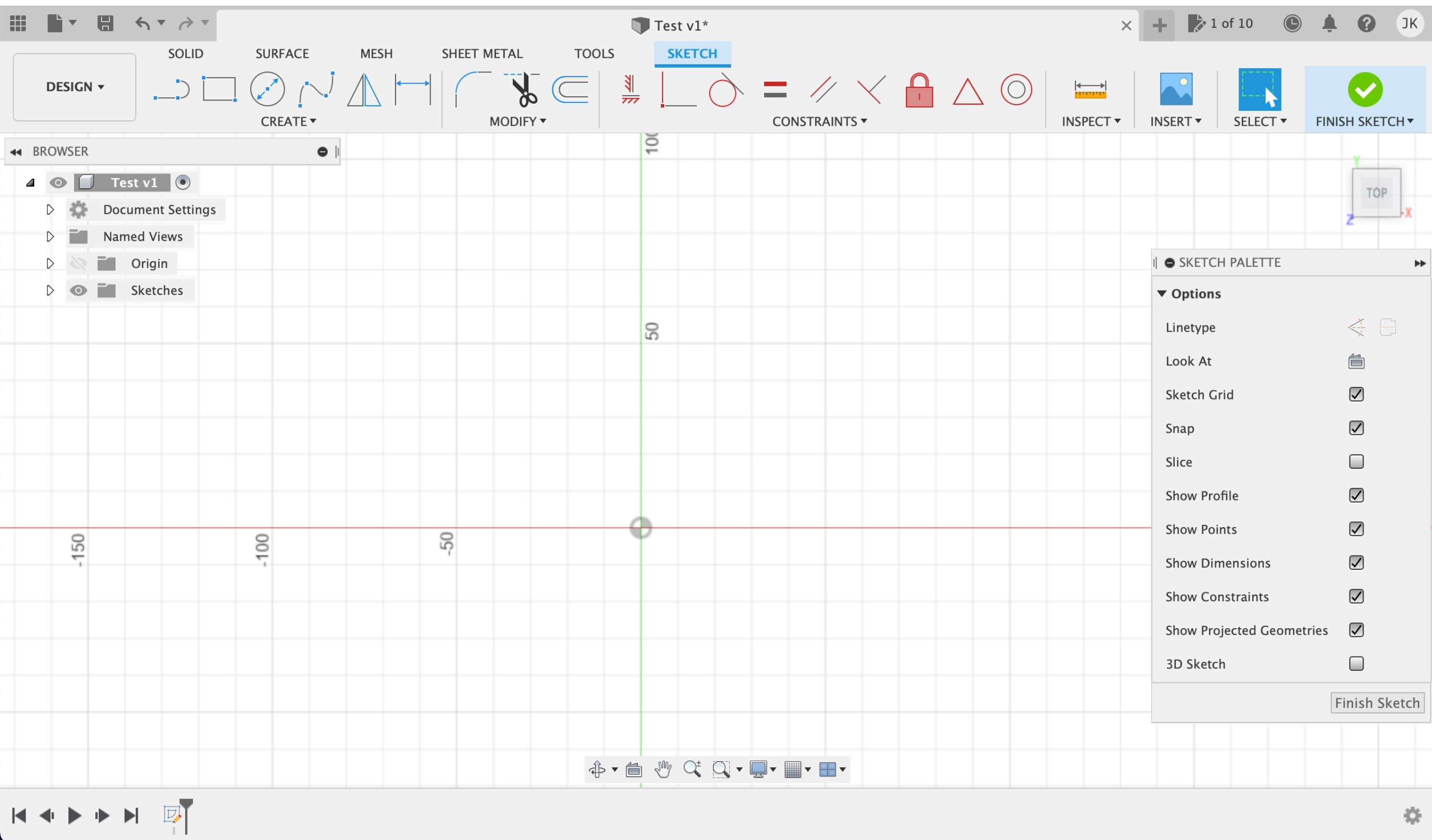


# Interface

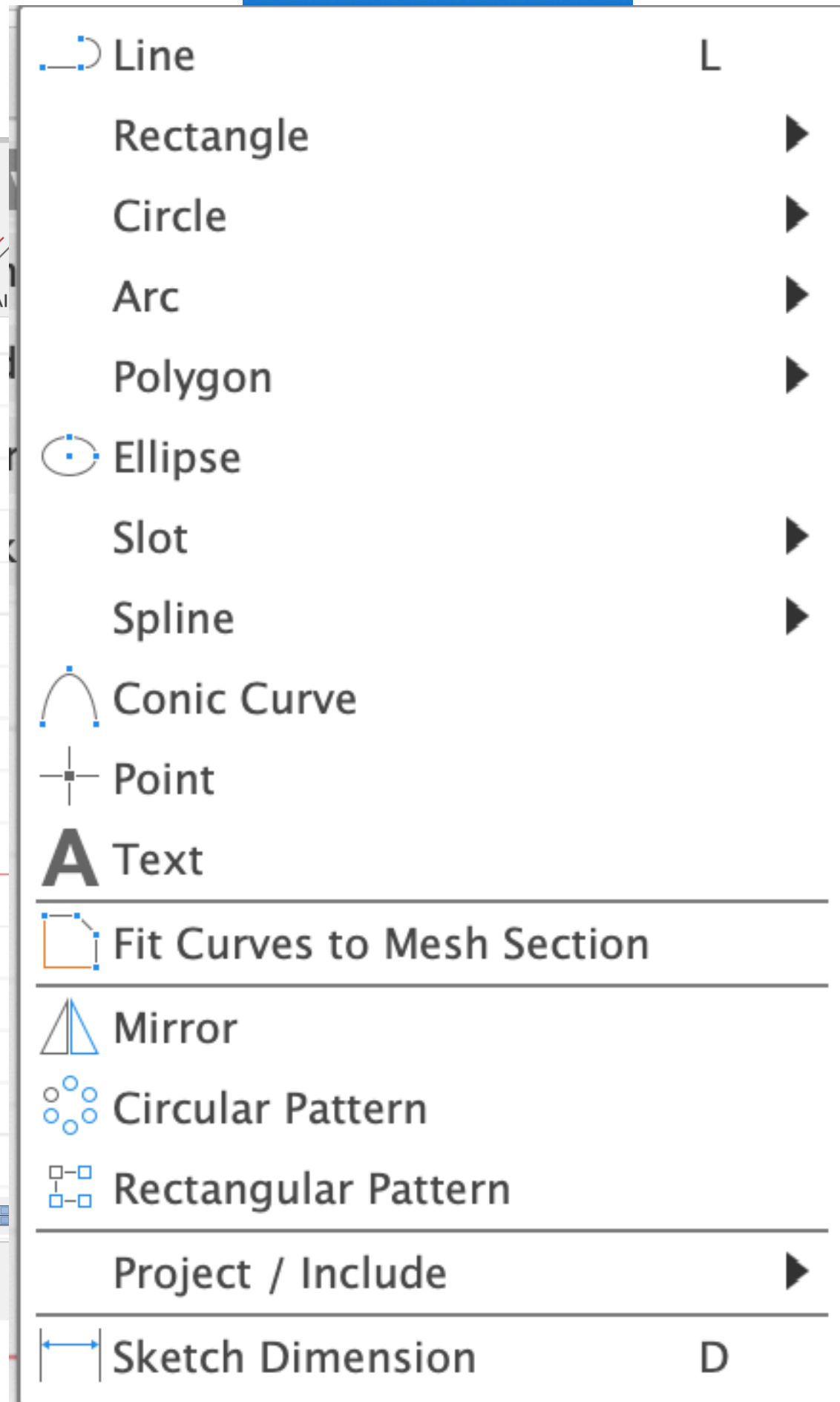
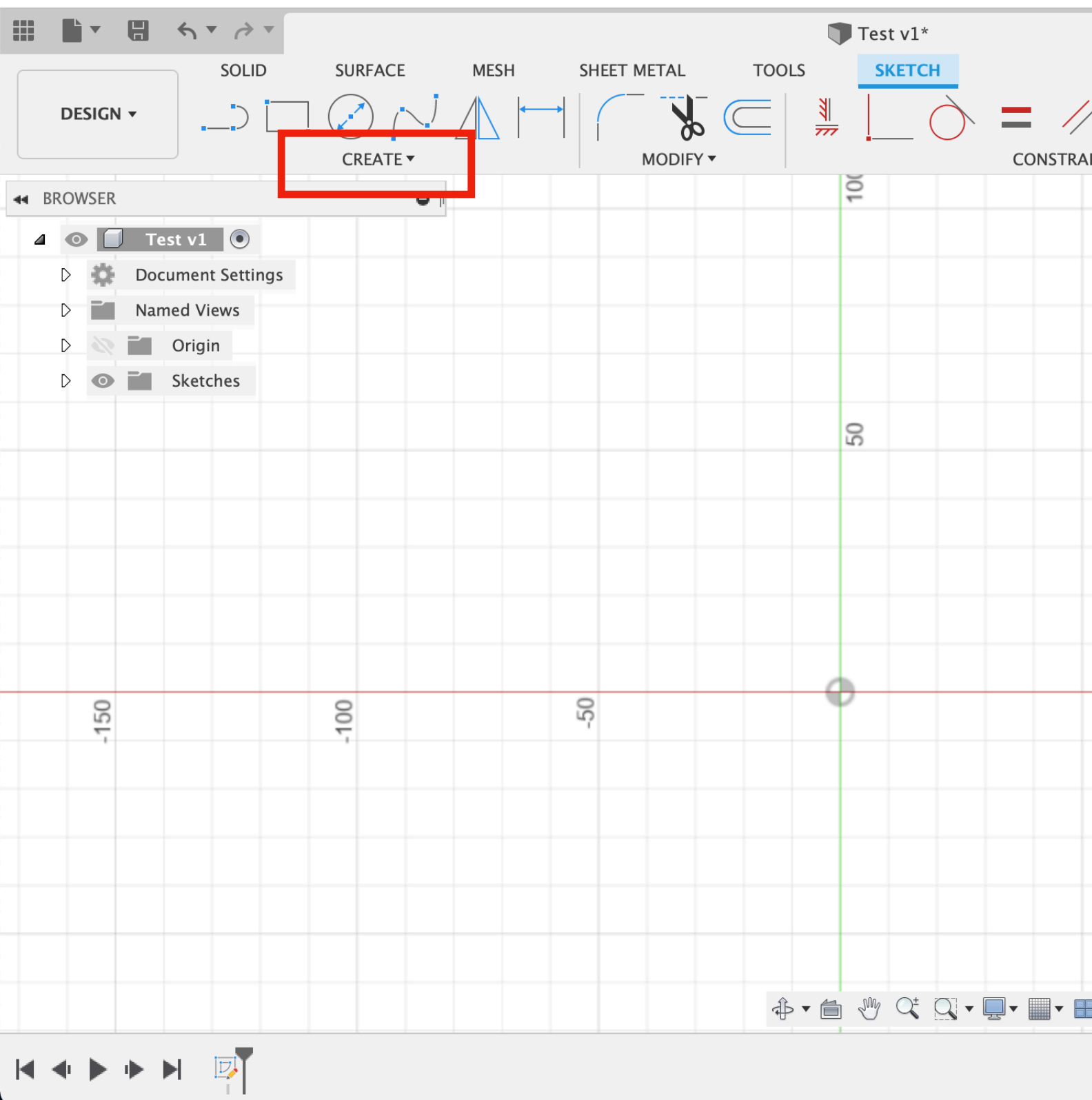




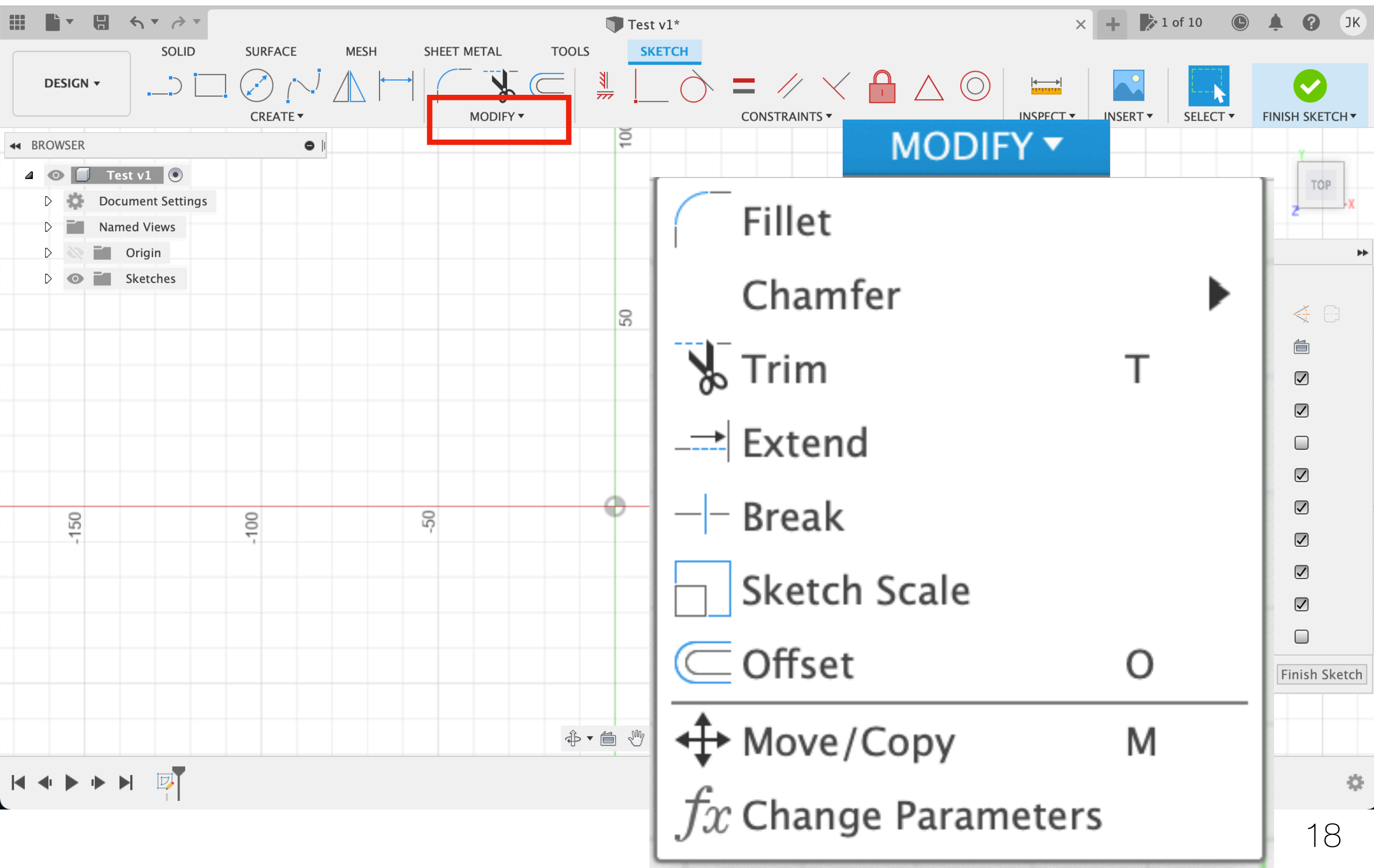
# Interface



# Interface

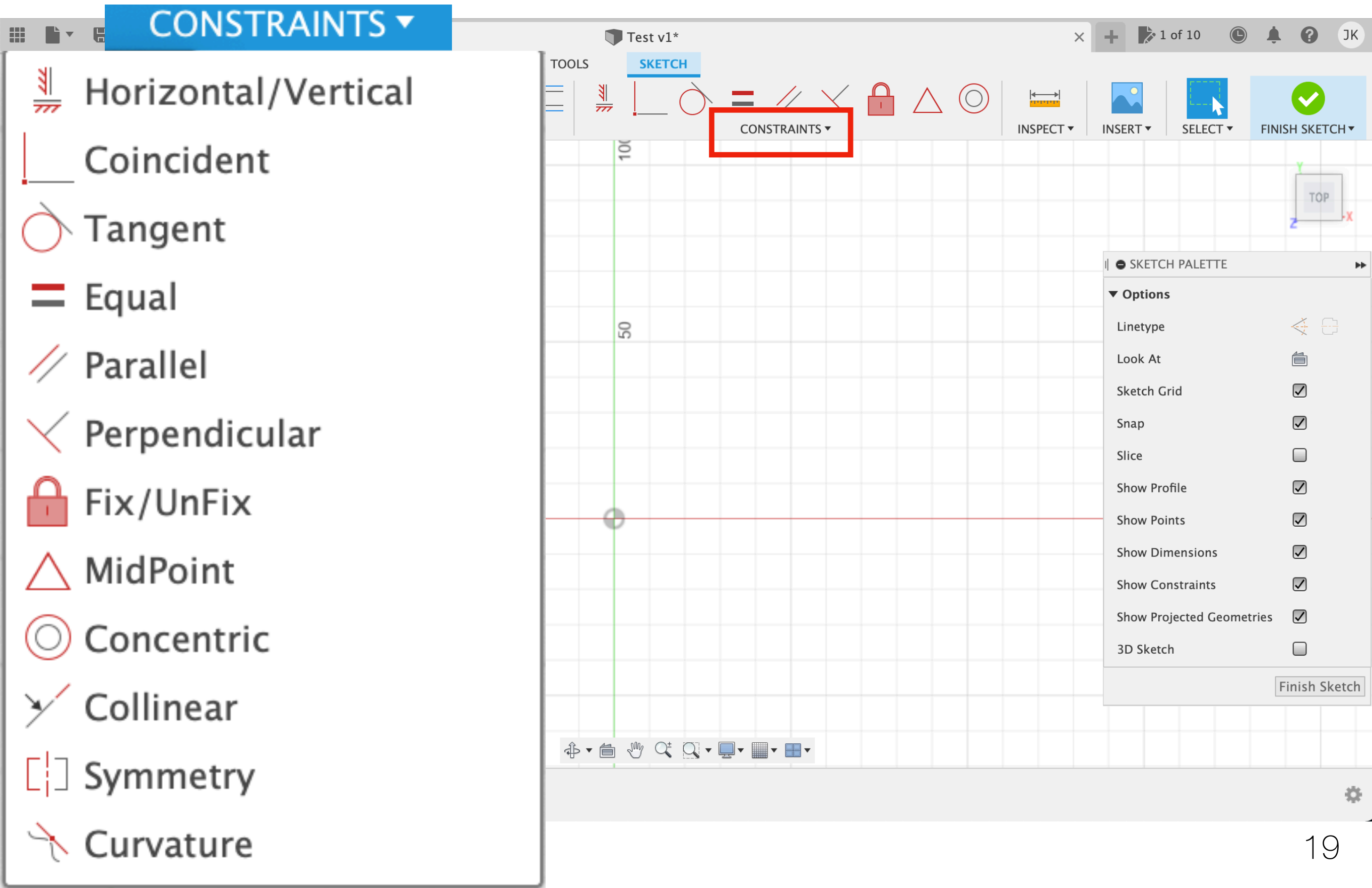


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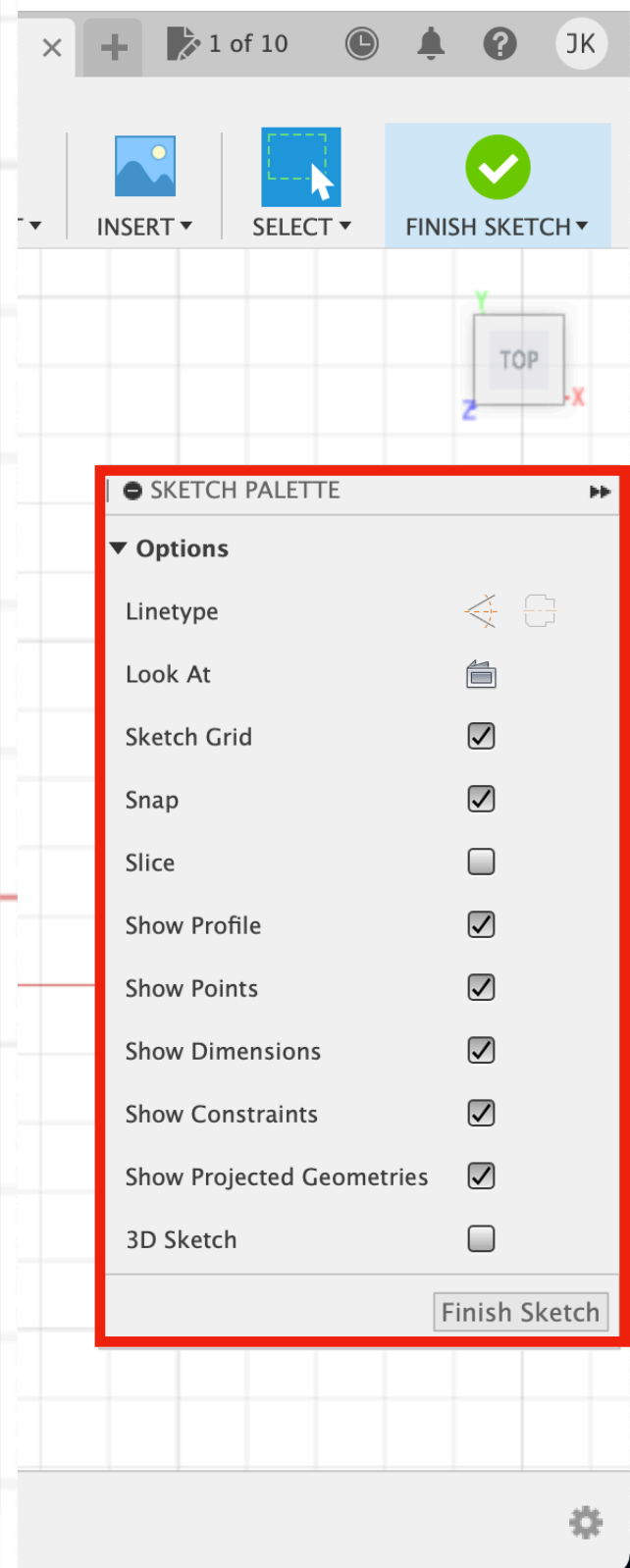
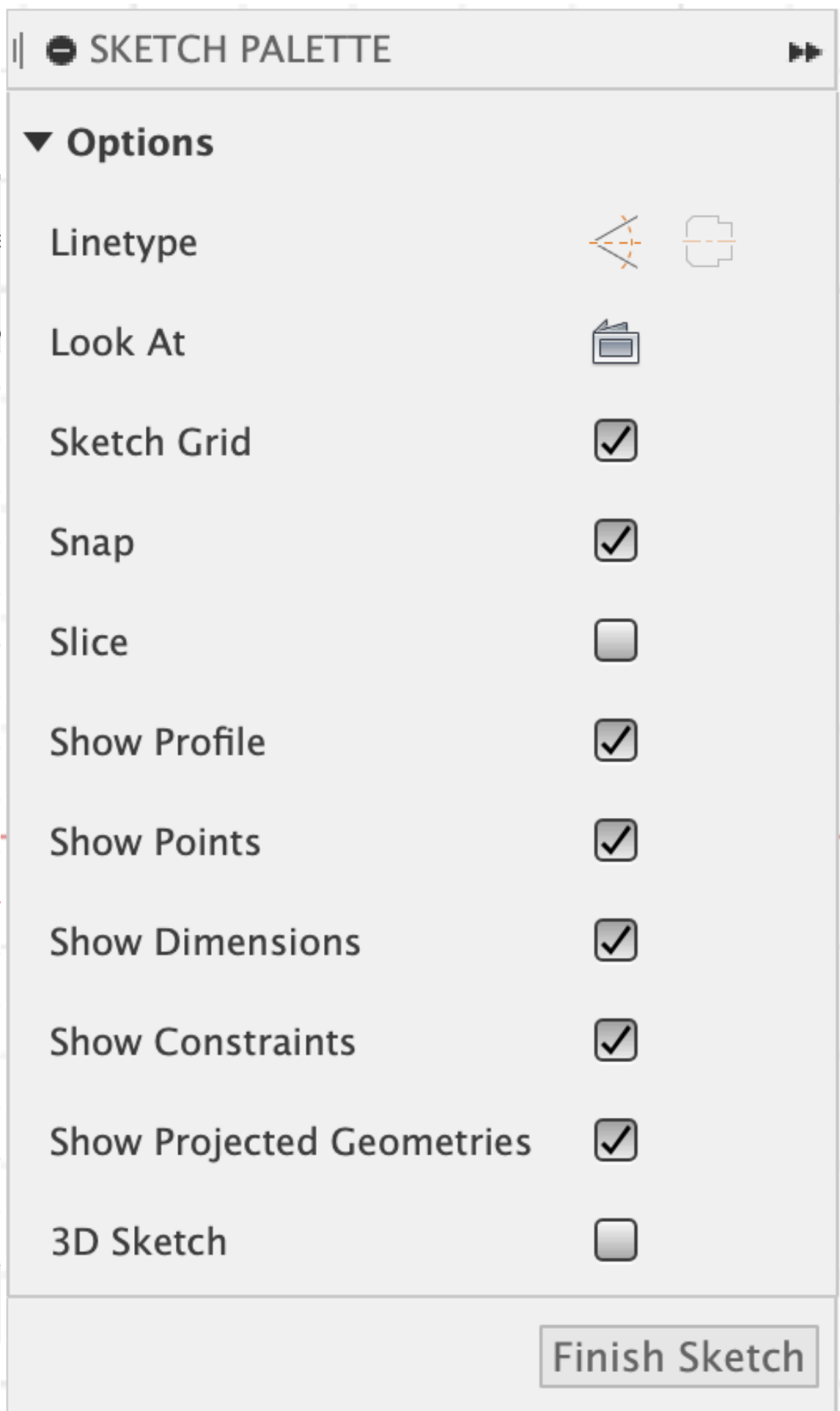
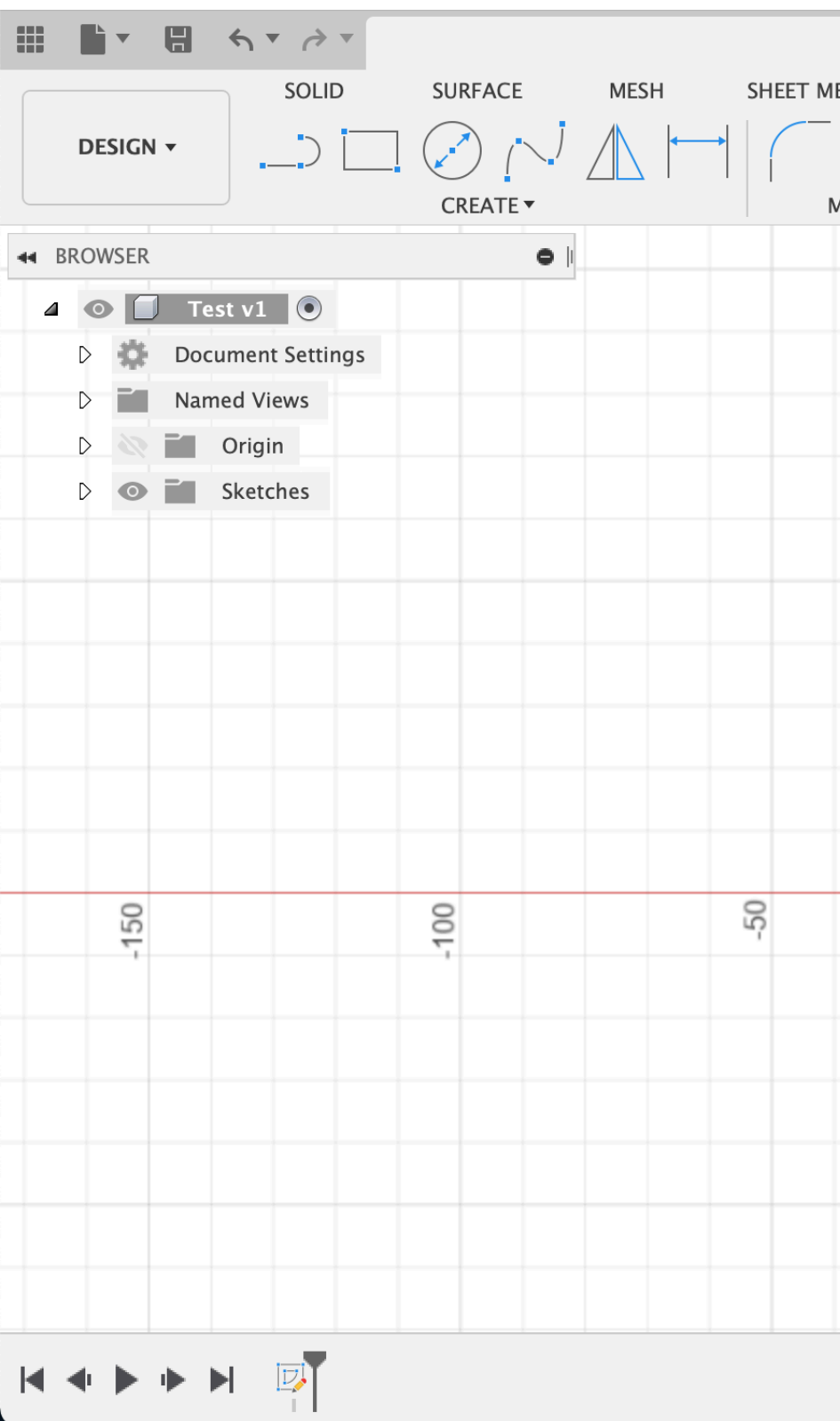




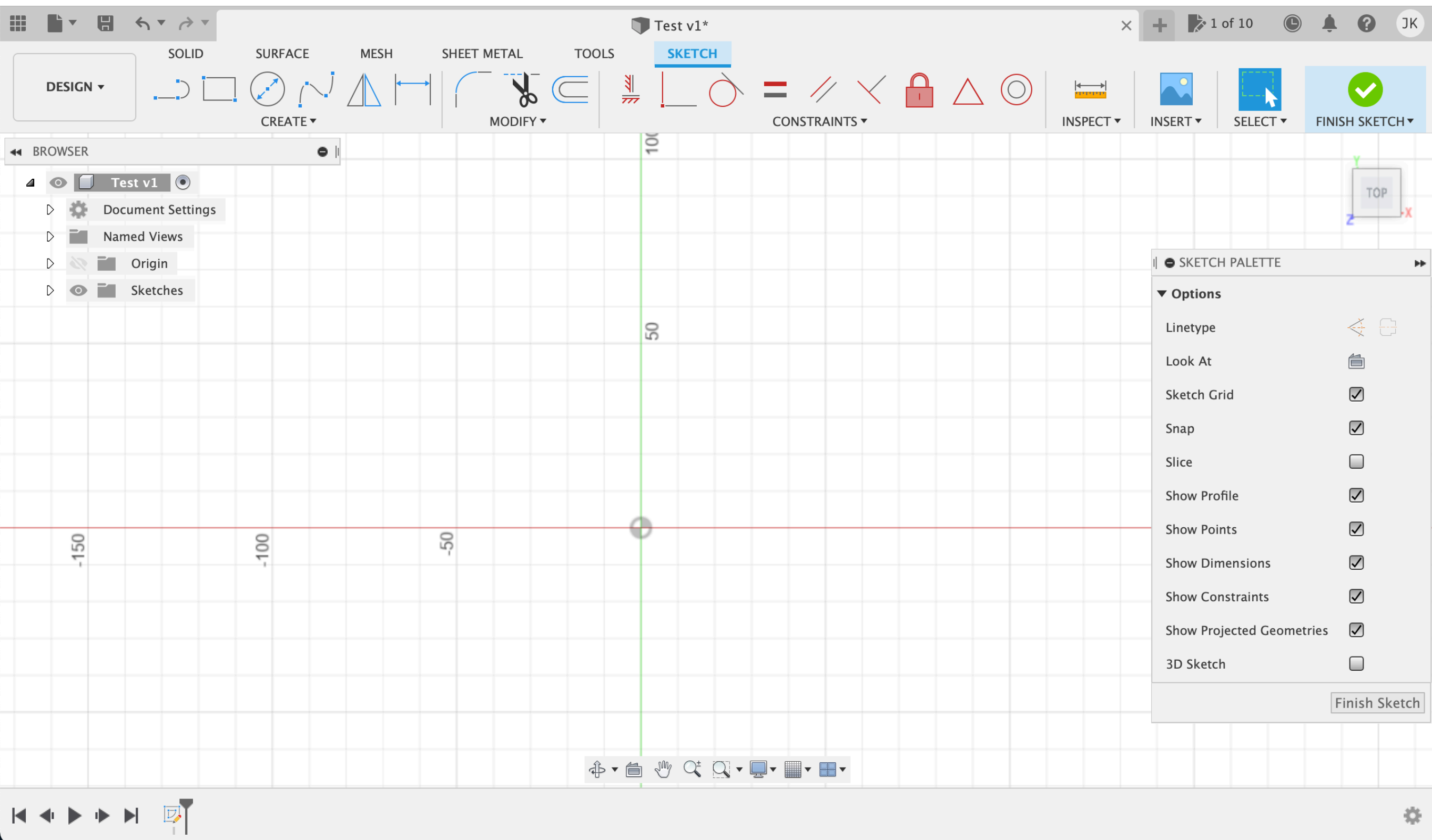
# Interface



# Interface



# Interface





# Practice

In Fusion 360, the triangle symbol  $\triangle$  corresponds to which constraint type?

A

B

C

D

- A) equilateral
- B) equiangular
- C) midpoint
- D) coincident

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